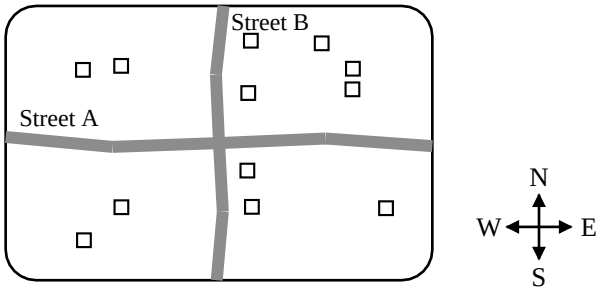
Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q12 (v1)

One street splits all the houses into two groups: north + south = all houses = east + west

How to think about it: one village, two streets

Street A goes from west to east. Street B goes from north to south. They cross in the middle of the village. Here is a village where you can see the houses:



- **Street A** cuts the village into a north part and a south part. Count: north 7, south 5. So the village has 7 + 5 = **12 houses**.
- **Street B** cuts the **same village** into a west part and an east part. Count: west 4, east 8. And 4 + 8 = **12 houses** again!

The big idea: every house is either north or south of street A, and every house is either west or east of street B. So north + south = all the houses = west + east.

Step by step: the contest question

In the contest the houses are **not** drawn. You only know: north 7, east 8, south 5. How many are west?

- **1. Use the two numbers of the same street.** North and south both belong to street A: 7 + 5 = 12 houses in the whole village.
- **2. Now use the other street.** East of B there are 8, so west of B there are 12 – 8 = **4**.

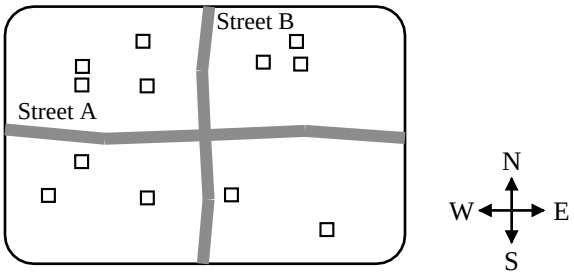
	street A	street B
one side	north: 7	east: 8
other side	south: 5	west: ?
all houses	12	12

Watch out

- You do not need to know which corner each house is in. You only need the total.
- Do not just add or subtract the numbers in the order you read them. First ask: which numbers belong to the **same** street?
- Check: west + east must give the same total as north + south (4 + 8 = 12 = 7 + 5).

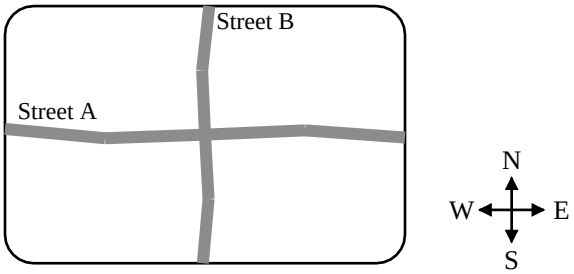
Warm-up: small steps first (no points)

W1. Look at the map. How many houses are there east of street B?



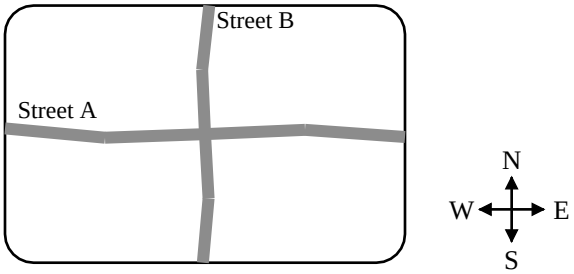
A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 12

W2. North of street A there are 5 houses. South of street A there are 10 houses. No house stands on a street. How many houses are there in the village?



A. 15 ***B.*** 16 ***C.*** 17 ***D.*** 18 ***E.*** 19

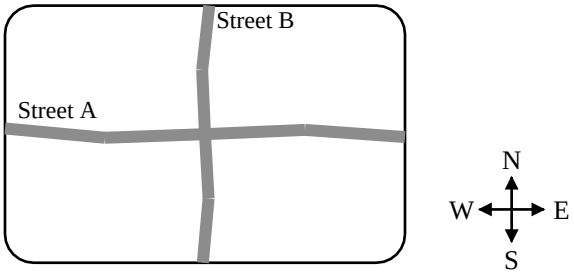
W3. There are 10 houses in the village. East of street B there are 7 houses. How many houses are there west of street B?



A. 3 ***B.*** 4 ***C.*** 7 ***D.*** 10 ***E.*** 17

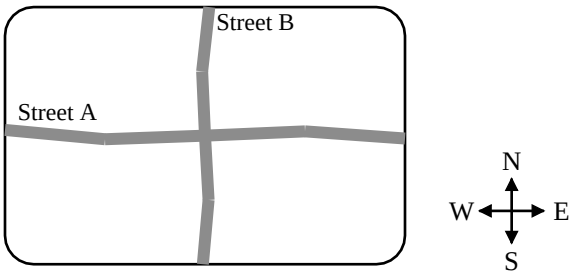
Part A: Each question is worth 3 points

1. North of street A there are 4 houses. East of street B there are 7 houses. South of street A there are 5 houses. How many houses are there west of street B?



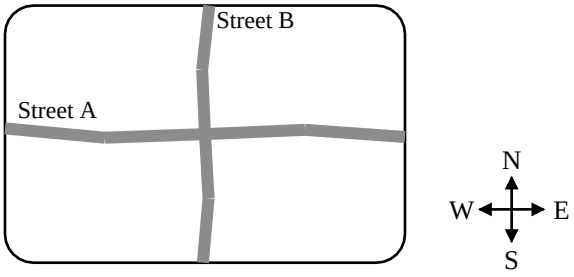
A. 0 ***B.*** 2 ***C.*** 5 ***D.*** 7 ***E.*** 16

2. North of street A there are 10 houses. West of street B there are 5 houses. South of street A there are 7 houses. How many houses are there east of street B?



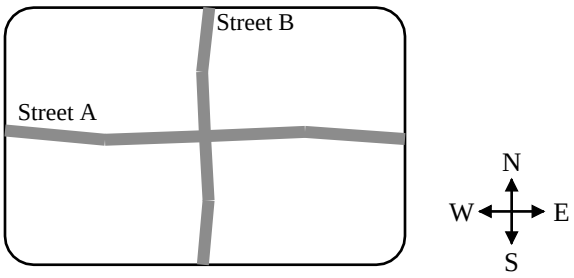
A. 5 ***B.*** 7 ***C.*** 10 ***D.*** 12 ***E.*** 22

3. North of street A there are 7 houses. East of street B there are 7 houses. South of street A there are 10 houses. No house stands on a street. How many houses are there in the village altogether?



A. 10 houses ***B.*** 14 houses ***C.*** 17 houses ***D.*** 19 houses ***E.*** 24 houses

4. North of street A there are 10 houses. West of street B there are 13 houses. South of street A there are 7 houses. No house stands on a street. How many houses are there in the village altogether?



A. 4 houses ***B.*** 16 houses ***C.*** 17 houses ***D.*** 20 houses ***E.*** 23 houses

Part B: Each question is worth 4 points

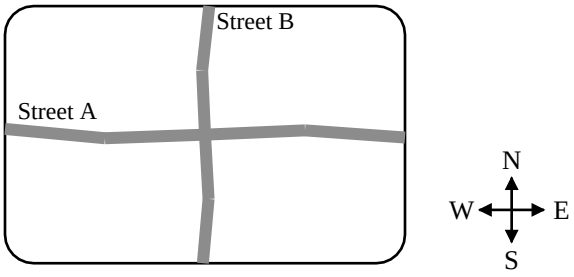
5. In a class, 11 children have a pet and 13 children have no pet. 12 children wear glasses. How many children do not wear glasses?

A. 1 ***B.*** 9 ***C.*** 10 ***D.*** 12 ***E.*** 24

6. In a class, 18 children have a sister and 22 children have no sister. 20 children can swim. How many children cannot swim?

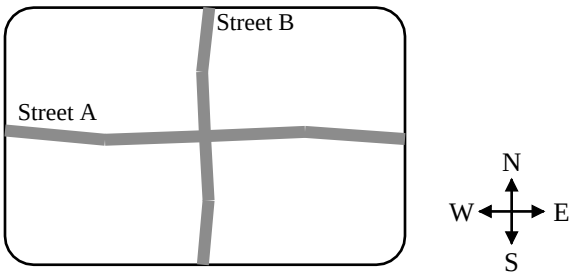
A. 2 ***B.*** 19 ***C.*** 20 ***D.*** 40 ***E.*** 60

7. North of street A there are 2 houses. South of street A there are 4 houses. West of street B there are just as many houses as east of street B. How many houses are there west of street B?



A. 3 houses ***B.*** 4 houses ***C.*** 5 houses ***D.*** 6 houses ***E.*** 7 houses

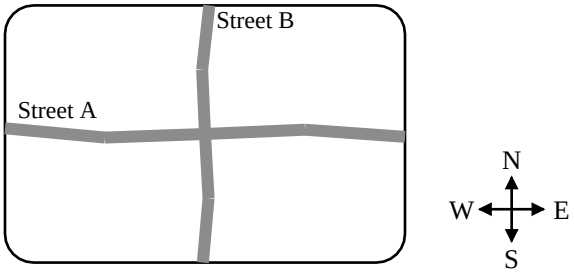
8. North of street A there are 9 houses. South of street A there are 11 houses. West of street B there are just as many houses as east of street B. How many houses are there east of street B?



A. 2 houses ***B.*** 7 houses ***C.*** 8 houses ***D.*** 9 houses ***E.*** 10 houses

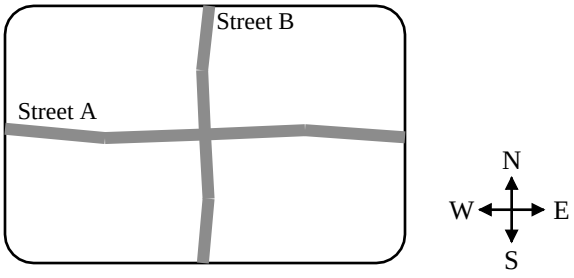
Part C: Each question is worth 5 points

9. North of street A there are 9 houses. South of street A there are 8 houses. East of street B there are 10 houses. 5 of the houses are north of street A **and** east of street B. How many houses are south of street A **and** west of street B?



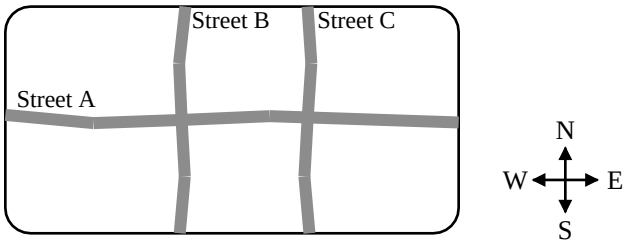
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 7 ***E.*** 8

10. North of street A there are 12 houses. South of street A there are 11 houses. East of street B there are 11 houses. 3 of the houses are north of street A **and** east of street B. How many houses are south of street A **and** west of street B?



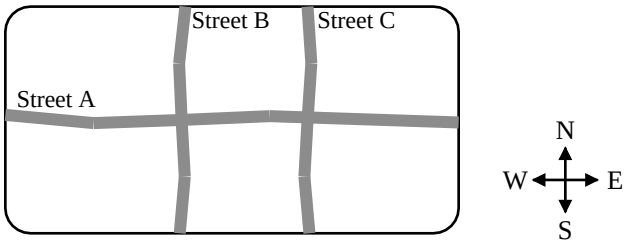
A. 1 ***B.*** 3 ***C.*** 9 ***D.*** 11 ***E.*** 12

11. Streets B and C both go from north to south (see map). North of street A there are 4 houses. South of street A there are 5 houses. West of street B there are 2 houses. East of street C there are 4 houses. How many houses are there between street B and street C?



A. 1 house ***B.*** 3 houses ***C.*** 5 houses ***D.*** 6 houses ***E.*** 7 houses

12. Streets B and C both go from north to south (see map). North of street A there are 16 houses. South of street A there are 6 houses. West of street B there are 8 houses. East of street C there are 8 houses. How many houses are there between street B and street C?



A. 0 houses ***B.*** 6 houses ***C.*** 10 houses ***D.*** 14 houses ***E.*** 16 houses

Answer Key

W1.	**A**	**W2.**	**A**	**W3.**	**A**	**1.**	**B**	**2.**	**D**	**3.**	**C**
4.	**C**	**5.**	**D**	**6.**	**C**	**7.**	**A**	**8.**	**E**	**9.**	**C**
10.	**B**	**11.**	**B**	**12.**	**B**						

Solutions

W1. A (5) — Count the little squares on the east side of street B: 5. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · w_count_on_map]

W2. A (15) — Every house is north or south of street A: 5 + 10 = 15. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · w_one_street]

W3. A (3) — Every house is east or west of street B: 10 – 7 = 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · w_other_side]

1. B (2) — Every house is either north or south of street A: 4 + 5 = 9 houses. Every house is also either east or west of street B: 9 – 7 = 2. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · cross_streets]

2. D (12) — Every house is either north or south of street A: 10 + 7 = 17 houses. Every house is also either west or east of street B: 17 – 5 = 12. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · cross_streets]

3. C (17 houses) — Every house is either north or south of street A, so the village has 7 + 10 = 17 houses. The east houses are already counted in these. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · whole_village]

4. C (17 houses) — Every house is either north or south of street A, so the village has 10 + 7 = 17 houses. The west houses are already counted in these. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · whole_village]

5. D (12) — There are 11 + 13 = 24 children in the class. 12 of them wear glasses, so the others do not wear glasses: 24 – 12 = 12. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · class_groups]

6. C (20) — There are 18 + 22 = 40 children in the class. 20 of them can swim, so the others cannot swim: 40 – 20 = 20. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · class_groups]

7. A (3 houses) — The village has 2 + 4 = 6 houses. Street B cuts them into two equal groups: 6 ÷ 2 = 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · same_both_sides]

8. E (10 houses) — The village has 9 + 11 = 20 houses. Street B cuts them into two equal groups: 20 ÷ 2 = 10. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · same_both_sides]

9. C (3) — The village has 9 + 8 = 17 houses, so 17 – 10 = 7 are west of street B. North-west there are 9 – 5 = 4 houses. So south-west there are 7 – 4 = 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · one_corner]

10. B (3) — The village has 12 + 11 = 23 houses, so 23 – 11 = 12 are west of street B. North-west there are 12 – 3 = 9 houses. So south-west there are 12 – 9 = 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · one_corner]

11. B (3 houses) — The village has 4 + 5 = 9 houses. Take away the houses west of B and east of C: 9 – 2 – 4 = 3 houses are left between the two streets. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · three_streets]

12. B (6 houses) — The village has 16 + 6 = 22 houses. Take away the houses west of B and east of C: 22 – 8 – 8 = 6 houses are left between the two streets. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · three_streets]